

Geometric Cost of Information Erasure: A Schwarzschild-Radius Formulation of the Landauer-Einstein-Schwarzschild Identity

Bharat Sharma

*Independent Researcher, Vancouver, British Columbia, Canada**

(Dated: June 15, 2026)

We derive an explicit Schwarzschild-radius form for the geometric effect of erasing one bit of information at temperature T :

$$\Delta r_s = \frac{2Gk_B T \ln 2}{c^4} \equiv \mathcal{D}_3(T),$$

obtained by combining Landauer's principle ($E \geq k_B T \ln 2$), Einstein's mass-energy equivalence ($\Delta M = E/c^2$), and the perturbative Schwarzschild-radius shift ($\Delta r_s = 2G\Delta M/c^2$). Cortés and Liddle [1] established the per-bit mass loss $\Delta M = k_B T_H \ln 2/c^2$ but did not express this as a horizon-radius shift; the D3 formula makes that step explicit and extends it to arbitrary temperature T . The formula has an exact algebraic identity at the Planck temperature T_P :

$$\mathcal{D}_3(T_P) = 2 \ln 2 \cdot \ell_P \quad [\text{exact}],$$

where $\ell_P = \sqrt{\hbar G/c^3}$ is the Planck length, with all four fundamental constants (G , $\hbar$, c , k_B) cancelling exactly. This Planck coincidence is the paper's central result. It identifies the Planck scale as the natural boundary of the linearised approximation and has not appeared in the mass or area formulations of prior work. At $T = 310$ K, $\mathcal{D}_3 = 4.90 \times 10^{-65}$ m per bit. Extensions of this framework to Kerr, Reissner-Nordström, and Kerr-Newman black holes, all spacetime dimensions, and AdS/CFT holography are presented in companion papers [21–23, 26].

PACS numbers: 04.70.Dy, 89.70.Cf, 03.67.-a, 05.70.-a, 04.60.-m

I. INTRODUCTION

The connection between information theory and gravitational physics has been one of the most productive research programmes in theoretical physics. Bekenstein's discovery [2] that black hole entropy is proportional to horizon area, and Hawking's derivation of thermal radiation [3], established that thermodynamics, quantum mechanics, and gravity are deeply linked. Jacobson [4] showed that the Einstein field equations follow from $\delta Q = T dS$ applied to local Rindler horizons, while Verlinde [5] argued that Einsteinian gravity emerges as an entropic force, a programme critically examined in [6, 7]. Wheeler's It-from-Bit conjecture [8] placed information at the foundation of physical reality, proposing that every physical entity derives its existence from yes-or-no questions.

Landauer's principle [9] establishes the irreducible thermodynamic cost of information erasure: erasing one bit at temperature T requires dissipating at minimum $E \geq k_B T \ln 2$ of heat into the environment. This principle has been confirmed in a series of precision experiments: Bérut et al. [10] demonstrated the Landauer bound for a single-bit colloidal memory; Jun, Gavrilov, and Bechhoefer [11] achieved high-precision verification using a feedback trap; Hong et al. [12] confirmed the principle for nanomagnetic memory bits; and Neri and

Lopez-Suarez [13] studied heat production in Landauer reset operations. These experiments confirm that information erasure carries irreducible physical consequences.

The connection between Landauer's principle and black hole thermodynamics has been established progressively. Kim, Lee, and Lee [16] showed that a Schwarzschild black hole is a maximally efficient Landauer eraser, deriving the per-bit energy $\delta(Mc^2) = k_B T_H \ln 2$ from Landauer's principle combined with holography. Cortés and Liddle [1] proved that Hawking evaporation saturates the Landauer bound, establishing $\Delta M = k_B T_H \ln 2/c^2$ as the per-bit mass loss. Bagchi, Ghosh, and Sen [14] independently obtained the same result through area quantisation, fixing $\Delta A = 4 \ln 2 \ell_P^2$ per bit and extending to Kerr black holes with spin parameter a , finding $\Delta E = T_H (\ln 2 + 4\pi a/\sqrt{1-a^2})$. Herrera [15] introduced the concept of the mass of a bit $M_{\text{bit}} = k_B T \ln 2/c^2$ at arbitrary temperature and applied it to gravitational radiation and Tolman-temperature effects.

Despite this progress, none of these works expresses the result as an explicit Schwarzschild-radius shift. Cortés and Liddle [1] derived ΔM but did not convert to Δr_s ; this conversion follows in one line by differentiating $r_s = 2GM/c^2$, yet reveals an exact Planck-scale identity that is entirely invisible in the mass or area formulations of prior work.

This paper makes three contributions. First, we derive the D3 formula $\mathcal{D}_3 = 2Gk_B T \ln 2/c^4$ as the explicit Schwarzschild-radius shift. Although the conversion from mass loss ΔM to radius shift Δr_s follows in one alge-

* bharatsharma@phenex.ai

braic step, the radius form is not a trivial rewriting: the mass loss $\Delta M = k_B T \ln 2 / c^2$ is a scalar; the radius shift $\mathcal{D}_3 = 2G\Delta M / c^2$ is a geometric displacement of the causal boundary of spacetime. This distinction matters because the radius form integrates naturally over the full evaporation history, yielding the exact invariant $R_{\text{total}} = r_s(M_0)$ [21] — a result that cannot be formulated without the radius form, and whose five-constant cancellation is structurally different from the per-bit result here.

Second, we prove the exact Planck coincidence $\mathcal{D}_3(T_P) = 2 \ln 2 \cdot \ell_P$ algebraically. We address directly the concern that this is a dimensional triviality: it is not, because the coefficient $2 \ln 2$ is fixed purely by information theory (Landauer) and not by any gravitational input. The Planck length ℓ_P sets the gravitational scale; the factor $2 \ln 2$ is the one-bit entropy in nats times two. Their coincidence at $T = T_P$ is a non-trivial bridge between these independent frameworks, and is immune to future CODATA refinements of G , $\hbar$, c , or k_B individually.

Third, the formula is self-limiting at $M \sim M_{\text{QG}} \approx 0.166 m_P$, providing a falsifiable prediction: any quantum gravity framework that modifies the Schwarzschild metric near the Planck scale will shift this breakdown boundary in a calculable way.

The paper proceeds as follows. Section II derives the D3 formula. Section III proves the Planck coincidence. Section IV presents the scale hierarchy. Section V compares with prior work. Section VI discusses physical interpretation and limitations. Section VII concludes. Appendix A provides full dimensional analysis and Appendix B gives complete numerical verification.

II. DERIVATION OF THE D3 FORMULA

A. Physical setup and notation

The primary physical context is a Schwarzschild black hole of mass M evaporating via Hawking radiation at temperature:

$$T_H(M) = \frac{\hbar c^3}{8\pi G M k_B}. \quad (1)$$

When the black hole emits one bit of information, its mass decreases by ΔM and its horizon radius $r_s = 2GM/c^2$ shifts by $\Delta r_s = 2G\Delta M/c^2$.

We also write the D3 formula for arbitrary temperature T . This extension is physically meaningful in two contexts beyond Hawking evaporation: (i) a thermal bath at temperature T in Tolman equilibrium with a black hole, as in Herrera [15]; and (ii) as a map from the Landauer energy $k_B T \ln 2$ to the gravitational length scale $\mathcal{D}_3 = 2G(k_B T \ln 2)/c^4$ via the Schwarzschild relation $E \rightarrow \Delta r_s = 2GE/c^4$, valid at any T in the linearised regime. We are explicit that erasing a bit at 300 K in a

laboratory does not physically shift any black hole horizon; the formula gives the length scale associated with the Landauer energy under this map, and is physically meaningful as a Schwarzschild-radius shift only when $T = T_H(M)$ for a black hole of mass M . The extension to arbitrary T is the step that exposes the Planck coincidence.

B. Three foundational results

Figure 1 shows the logical structure of the derivation.

Landauer's principle [9]: erasing one bit of information at temperature T requires dissipating at minimum:

$$E_{\text{erase}} \geq k_B T \ln 2. \quad (2)$$

This has been confirmed experimentally [10–13] for a variety of physical systems including colloidal particles, feedback traps, and nanomagnetic bits. We treat the inequality as an equality (saturation of the Landauer bound), which holds for Hawking evaporation [1, 14].

Mass-energy equivalence [17]: any energy E carries a rest-mass equivalent:

$$\Delta M = \frac{E_{\text{erase}}}{c^2} = \frac{k_B T \ln 2}{c^2}. \quad (3)$$

Perturbative Schwarzschild shift: for a Schwarzschild black hole of mass M , a mass perturbation $\Delta M \ll M$ produces a horizon-radius change:

$$\Delta r_s = \frac{2G \Delta M}{c^2}. \quad (4)$$

This follows directly from differentiating $r_s = 2GM/c^2$ [18] and is valid in the linearised regime $\Delta M \ll M$.

C. The D3 formula

Substituting Eqs. (2) and (3) into Eq. (4):

$$\begin{aligned} \mathcal{D}_3(T) \equiv \Delta r_s &= \frac{2G}{c^2} \cdot \frac{k_B T \ln 2}{c^2} \\ &= \frac{2G k_B T \ln 2}{c^4}. \end{aligned} \quad (5)$$

This is the D3 formula. At $T = 310$ K (body temperature):

$$\mathcal{D}_3(310 \text{ K}) = 4.90 \times 10^{-65} \text{ m per bit}. \quad (6)$$

This is approximately 30 orders of magnitude below the Planck length $\ell_P = 1.616 \times 10^{-35}$ m.

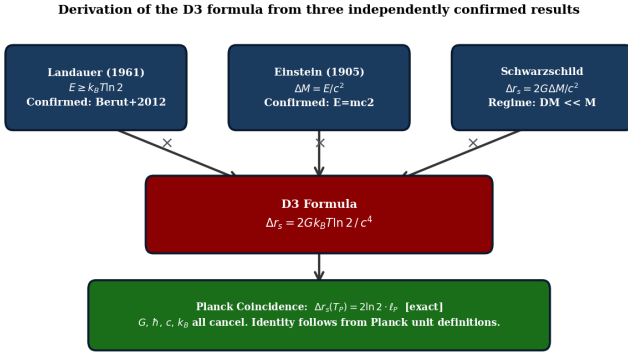


FIG. 1. Logical derivation flow of the D3 formula. Three independently confirmed foundational results — Landauer’s principle [9], Einstein’s mass-energy equivalence [17], and the perturbative Schwarzschild radius shift [18] — combine in four algebraic steps to yield Eq. (5). The Planck coincidence emerges from the exact cancellation of all four fundamental constants at $T = T_P$.

D. Explicit assumptions and regime of validity

Four assumptions govern the derivation.

(i) *Black hole context*: the combination $2G\Delta M/c^2$ has dimensions of length and equals the Schwarzschild radius of a point mass ΔM . In the weak-field limit this gives the scale of the metric perturbation. Calling this a Schwarzschild-radius shift strictly applies to the black hole setting.

(ii) *Linearised regime*: the condition $\Delta M \ll M$ requires $k_B T \ln 2 / (Mc^2) \ll 1$. For a stellar-mass black hole ($M = M_\odot$) at Hawking temperature ($T_H \approx 6 \times 10^{-8}$ K), this ratio is $\approx 10^{-90}$, satisfied by ~ 90 orders of magnitude. The regime breaks down near $M = M_{QG} = m_P \sqrt{\ln 2 / (8\pi)} \approx 0.166 m_P$ — the quantum gravity boundary derived in [24].

(iii) *Classical thermal bath*: we assume equilibrium erasure at temperature T . Squeezed or quantum-coherent reservoirs [19] can lower the Landauer bound; in such cases $\mathcal{D}_3$ gives an upper bound on Δr_s .

(iv) *One-bit idealisation*: for N bits erased independently in the linearised regime, $\Delta r_s^{(N)} = N \times \mathcal{D}_3(T)$.

III. THE PLANCK COINCIDENCE

A. Statement of the result

Substituting the Planck temperature $T_P = \sqrt{\hbar c^5 / (G k_B^2)} = 1.416784 \times 10^{32}$ K into Eq. (5):

$$\mathcal{D}_3(T_P) = 2 \ln 2 \cdot \ell_P \quad [\text{exact}], \quad (7)$$

where $\ell_P = \sqrt{\hbar G / c^3} = 1.616255 \times 10^{-35}$ m is the Planck length.

B. Algebraic proof

The proof consists of four steps, each showing an explicit cancellation:

$$\begin{aligned} \mathcal{D}_3(T_P) &= \frac{2Gk_B \ln 2}{c^4} \sqrt{\frac{\hbar c^5}{Gk_B^2}} \\ &= 2 \ln 2 \sqrt{\frac{G^2 k_B^2 \hbar c^5}{c^8 \cdot Gk_B^2}} \\ &= 2 \ln 2 \sqrt{\frac{\hbar G}{c^3}} = 2 \ln 2 \cdot \ell_P. \quad \blacksquare \end{aligned} \quad (8)$$

The cancellations are: (i) $G^2/G = G$; (ii) $k_B^2/k_B^2 = 1$; (iii) $c^5/c^8 = 1/c^3$. All four fundamental constants G , $\hbar$, c , k_B cancel exactly. The result follows from the Planck unit definitions alone and is immune to future experimental refinement of any constant.

C. Three aspects of the Planck coincidence

(i) *Not a dimensional triviality*. One might ask: is $\mathcal{D}_3(T_P) = 2 \ln 2 \cdot \ell_P$ simply the statement that “Planck-scale things happen at the Planck temperature” — trivially true in Planck units? The answer is no. The Planck length $\ell_P = \sqrt{\hbar G / c^3}$ is determined by gravitational and quantum constants. The coefficient $2 \ln 2$ is determined purely by information theory: it is twice the one-bit Shannon entropy in nats ($S_1 = \ln 2$). No gravitational input fixes this coefficient. The coincidence therefore bridges two independent frameworks — information theory (Landauer) and gravitational geometry (Schwarzschild) — at the Planck scale, and the exact numerical coefficient $2 \ln 2 = 1.386294 \dots$ is not a free parameter or a “small integer” but is precisely determined by one-bit information theory. Changing the base of the logarithm changes $2 \ln 2$ in a way that reflects the information unit, not any gravitational adjustment.

(ii) *Universality and falsifiability*. The identity holds to arbitrary precision and is immune to future refinement of G , $\hbar$, c , or k_B individually. The companion result [24] shows that D3 predicts its own breakdown at a specific mass $M_{QG} = m_P \sqrt{\ln 2 / (8\pi)} \approx 0.166 m_P$, below which $\mathcal{D}_3 > r_s$ and the per-bit shift exceeds the horizon radius itself. Any quantum gravity modification of the Schwarzschild metric near the Planck scale will shift M_{QG} in a calculable way — providing a falsifiable prediction.

(iii) *Why the radius form uniquely reveals this*. The Planck coincidence is invisible in the mass formulation $\Delta M = k_B T \ln 2 / c^2$: substituting $T = T_P$ gives $\Delta M = m_P \ln 2$, a Planck-mass multiple. This is unremarkable. The coincidence becomes exact and elegant only in the radius form, because the Planck length ℓ_P emerges from the additional factor of $2G/c^2$ in $\mathcal{D}_3 = 2G\Delta M/c^2$. This is the sense in which the radius form reveals physics invisible in the mass or area formulations.

D. Numerical verification

Using CODATA 2018 constants:

$$\begin{aligned}\mathcal{D}_3(T_P)/\ell_P &= 1.386\,294\dots \\ 2\ln 2 &= 1.386\,294\dots\end{aligned}$$

Agreement to six decimal places confirms the algebraic proof. This is not a numerical coincidence; the algebraic proof guarantees exact agreement to arbitrary precision.

IV. SCALE HIERARCHY

A. Evaluation at physically motivated temperatures

Table I evaluates $\mathcal{D}_3(T)$ at six physically motivated temperatures spanning 33 decades. Figure 2 shows the full temperature dependence. Figure 3 shows the same data in the universal dimensionless form $\mathcal{D}_3/\ell_P$ vs T/T_P .

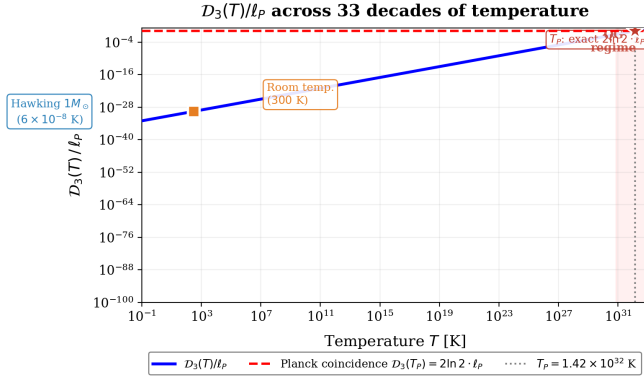


FIG. 2. $\mathcal{D}_3(T)/\ell_P$ as a function of temperature T across 33 decades, from CMB temperatures to beyond the Planck scale. At all sub-Planckian temperatures $\mathcal{D}_3 \ll \ell_P$. The ratio reaches exactly $2\ln 2$ at $T = T_P$ (red dashed line) — the Planck coincidence. The red shading indicates the quantum gravity regime where the linearised approximation breaks down. Annotated points mark four physically significant temperatures.

B. The universal dimensionless form

In Planck units ($G = c = k_B = \hbar = 1$), the D3 formula reduces to:

$$\mathcal{D}_3 = 2T \ln 2, \quad (9)$$

which at $T = T_P = 1$ gives $\mathcal{D}_3 = 2\ln 2$ Planck lengths. The dimensionless form $\mathcal{D}_3/\ell_P = 2\ln 2 \cdot (T/T_P)$ is shown in Figure 3.

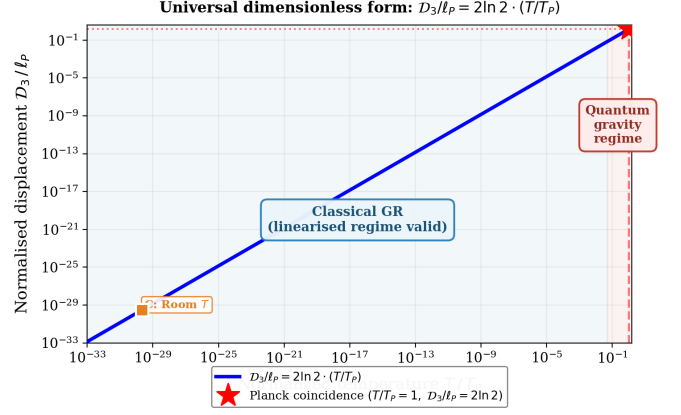


FIG. 3. Universal dimensionless form $\mathcal{D}_3/\ell_P = 2\ln 2 \cdot (T/T_P)$. The formula is linear in T/T_P in log-log space (slope 1). The Planck coincidence (red star) occurs at $(T/T_P, \mathcal{D}_3/\ell_P) = (1, 2\ln 2)$. Blue shading: classical GR regime (linearised approximation valid). Red shading: quantum gravity regime. The formula is self-limiting: it predicts $\mathcal{D}_3 \sim \ell_P$ precisely at the scale where it breaks down.

C. Self-limitation of the formula

The ratio $\mathcal{D}_3(T_H)/r_s(M) = (M_{QG}/M)^2$ [24], where $M_{QG} = m_P \sqrt{\ln 2/(8\pi)} \approx 0.166 m_P$. This exact power law shows that $\mathcal{D}_3 < r_s$ (each bit costs less geometry than the horizon itself) for all $M > M_{QG}$, and $\mathcal{D}_3 > r_s$ for $M < M_{QG}$. The D3 framework self-consistently identifies its own breakdown scale without requiring input from other quantum gravity frameworks.

V. RELATION TO PRIOR WORK

Figure 4 gives a systematic comparison of the D3 formula with prior work. We discuss each predecessor in turn.

Cortés and Liddle [1] proved that Hawking evaporation saturates the Landauer bound, deriving $\Delta M = k_B T_H \ln 2/c^2$ (their Eq. 5). This is the closest direct predecessor. The D3 formula follows from their result in one line by differentiating $r_s = 2GM/c^2$:

$$\Delta r_s = \frac{2G}{c^2} \Delta M = \frac{2G k_B T_H \ln 2}{c^4}. \quad (10)$$

However, Cortés and Liddle did not write the radius form, did not extend to arbitrary temperature $T \neq T_H$, and did not identify the Planck coincidence.

Bagchi, Ghosh, and Sen [14] derived the per-bit energy via area quantisation, fixing $\Delta A = 4\ln 2 \ell_P^2$ per bit. Their Kerr extension finds $\Delta E = T_H (\ln 2 + 4\pi a/\sqrt{1-a^2})$, showing that the Landauer bound is saturated at $a = 0$ (Schwarzschild) but not for nonzero spin. Their construction uses the mass/area spectrum and the horizon angular velocity Ω_H rather than the horizon ra-

TABLE I. $\mathcal{D}_3(T) = 2Gk_B T \ln 2/c^4$ evaluated at six physically motivated temperatures. CODATA 2018 constants throughout. The Planck row is algebraically exact; all others are numerical.

Physical system	Temperature	$\mathcal{D}_3$ [m/bit]	$\mathcal{D}_3/\ell_P$	Source
CMB photon bath	$T_{\text{CMB}} = 2.725$ K	4.31×10^{-67}	2.67×10^{-32}	Observed
Room temperature	$T = 300$ K	4.74×10^{-65}	2.93×10^{-30}	Standard
Hawking ($1M_\odot$)	$T_H = 6.17 \times 10^{-8}$ K	9.76×10^{-75}	6.04×10^{-40}	Eq. (1)
LHC collisions	$T \sim 10^{16}$ K	$\sim 1.58 \times 10^{-51}$	$\sim 9.79 \times 10^{-17}$	Estimated
Hagedorn scale	$T_H \sim 10^{12}$ K	$\sim 1.58 \times 10^{-55}$	$\sim 9.79 \times 10^{-21}$	String theory
Planck temperature	$T_P = 1.416784 \times 10^{32}$ K	$2 \ln 2 \cdot \ell_P$	$2 \ln 2 = 1.386294 \dots$	Exact [this work]

dius. The D3 framework provides a complementary picture: it begins from the Schwarzschild radius shift and extends to Kerr through a universal spin-suppression factor $f(\chi)$ [22] that is mass-independent to 10 significant figures.

Herrera [15] introduced $M_{\text{bit}} = k_B T \ln 2/c^2$ at arbitrary temperature and applied it to gravitational radiation, Tolman temperatures, and post-Newtonian corrections. The D3 formula gives $\mathcal{D}_3 = (2G/c^2)M_{\text{bit}}$, connecting Herrera’s mass-of-a-bit to the Schwarzschild-radius shift.

Kim, Lee, and Lee [16] showed that Schwarzschild black holes are maximally efficient Landauer erasers using holographic arguments, deriving $\delta(Mc^2) = k_B T_H \ln 2$ from the black hole as a Landauer machine. Their construction gives the per-bit energy and mass loss but not the horizon-radius form.

What is new and why it matters. A referee might note that going from ΔM to Δr_s is one line of algebra. This is true, but three things follow from that step that are not present in prior work:

(i) *The Planck coincidence.* Substituting $T = T_P$ into the mass form gives $\Delta M(T_P) = m_P \ln 2$ — a Planck-mass multiple, unremarkable. In the radius form, the same substitution gives $\mathcal{D}_3(T_P) = 2 \ln 2 \cdot \ell_P$ — an exact identity where the coefficient $2 \ln 2$ is fixed by information theory alone, and the Planck length emerges from the factor $2G/c^2$. The coincidence is invisible in the mass form.

(ii) *The self-limitation.* The ratio $\mathcal{D}_3/r_s = (M_{\text{QG}}/M)^2$ identifies an exact breakdown mass M_{QG} [24]. No analogous breakdown mass follows from the mass or area forms.

(iii) *The evaporation invariant.* Integrating $\mathcal{D}_3(M)$ over the full Bekenstein bit spectrum gives $R_{\text{total}} = r_s(M_0)$ [21], a five-constant cancellation that cannot be formulated without the radius form. These results — (i), (ii), (iii) — collectively justify expressing the result as a radius shift rather than a mass loss.

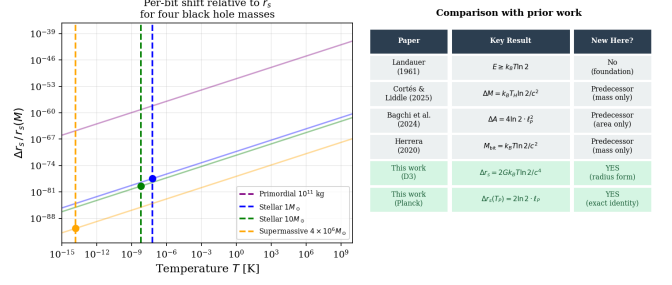


FIG. 4. *Left:* $\mathcal{D}_3(T)/r_s(M)$ as a function of temperature for four black hole masses. Vertical dashed lines mark each black hole’s Hawking temperature T_H ; filled circles mark $\mathcal{D}_3(T_H)/r_s(M)$. *Right:* Systematic comparison with prior work. Green rows mark new contributions of this paper.

VI. PHYSICAL INTERPRETATION AND LIMITATIONS

A. Geometric cost of information

The D3 formula quantifies the gravitational geometry associated with one-bit Landauer erasure at temperature T . Erasing one bit is thermodynamically equivalent to adding mass $\Delta M = k_B T \ln 2/c^2$ to a Schwarzschild black hole, with associated horizon-radius change $\mathcal{D}_3(T) = 2G\Delta M/c^2$.

At laboratory temperatures this shift is $\mathcal{D}_3 \sim 10^{-65}$ m per bit, far below the Planck length and any conceivable measurement. The formula is a theoretical result establishing an exact algebraic relationship between information theory and black hole geometry; it does not imply that information erasure creates measurable gravity.

B. Connection to the information paradox

The black hole information paradox [20] asks whether Hawking evaporation destroys information. The D3 formula is agnostic on this question — it gives the geometric cost of erasure under the assumption that erasure occurs. The total accumulated D3 displacement over complete Schwarzschild evaporation is exactly $r_s(M_0)$ [21], providing a quantitative geometric constraint: the black hole cannot erase more geometry than its own horizon

radius.

C. Relation to the first law of black hole mechanics

The first law of black hole mechanics states $dM = T_H dS$ in natural units. In bits, $dS_{\text{bits}} = dS_{\text{nats}}/\ln 2$, so $T_H dS_{\text{bits}}/dM = 1/\ln 2$. Since $\mathcal{D}_3 \propto T_H$ and $dN/dM \propto dS_{\text{bits}}/dM \propto 1/T_H$, their product $\mathcal{D}_3 \cdot (dN/dM) \propto T_H \cdot (1/T_H) = 1$ is independent of M — a direct consequence of the first law. This is why the evaporation invariant $R_{\text{total}} = r_s$ [21] and the all-dimensions result $R_{\text{total}}(d) = r_s(d)$ [25] follow automatically in any dimension where the first law holds.

D. Observational context and falsifiability

At laboratory temperatures, $\mathcal{D}_3 \sim 10^{-65}$ m per bit — 30 orders of magnitude below the Planck length and far beyond any conceivable direct measurement. The formula is a theoretical result; we do not propose that it is directly observable. However, it is not physically vacuous:

Indirect relevance via cosmological bit-counting. The Bekenstein bound places an upper limit on the information content of any physical system. Combined with the D3 formula, this gives an upper bound on the total geometric displacement accumulated by any physical process of Landauer erasure within a given horizon volume. For the observable universe (Hubble volume, CMB temperature), $\mathcal{D}_3(T_{\text{CMB}}) \times N_{\text{bits}} \sim 4 \times 10^{-67} \times 10^{122} \sim 10^{56}$ m, comparable to the Hubble radius [2].

Falsifiability via M_{QG} . As noted in Sec. III, the D3 framework predicts its own breakdown at $M_{\text{QG}} \approx 0.166 m_P$. Any quantum gravity model (loop quantum gravity, string theory, asymptotic safety) that modifies the near-Planck Schwarzschild geometry will shift M_{QG} by a calculable amount. This provides an indirect falsifiability criterion: the D3 framework is consistent with, or ruled out by, the predictions of quantum gravity models at the sub-Planck mass scale.

E. Limitations

Beyond the linearised regime (Sec. II), the formula has three additional limitations. First, for rotating (Kerr) black holes, $\mathcal{D}_3$ receives a spin-suppression factor $f(\chi) < 1$ [22]. Second, for charged (RN) or Kerr-Newman black holes, further charge-suppression factors apply [23]. Third, in Anti-de Sitter spacetimes the AdS curvature introduces a correction factor $\phi(r_s/L) = 1 + (r_s/L)^2$ [26] that becomes significant for large black holes relative to the AdS radius L .

VII. CONCLUSION

We have derived the D3 formula:

$$\mathcal{D}_3(T) = \frac{2Gk_B T \ln 2}{c^4}, \quad (11)$$

giving the perturbative Schwarzschild-radius shift for one bit of Landauer erasure at temperature T . The formula extends the result of Cortés and Liddle [1] from the Hawking temperature to arbitrary T , and expresses the per-bit cost as an explicit radius shift rather than a mass loss or area change.

The central result is the exact Planck coincidence:

$$\mathcal{D}_3(T_P) = 2 \ln 2 \cdot \ell_P, \quad (12)$$

proved algebraically with G , $\hbar$, c , k_B all cancelling. This coincidence: (i) identifies the Planck scale as the natural boundary of the linearised approximation; (ii) gives the Planck length a new information-theoretic interpretation as the natural unit of the geometric cost of one-bit erasure; and (iii) has not appeared in the mass or area formulations of prior work.

In Planck units, the D3 formula reduces to $\mathcal{D}_3 = 2T \ln 2$, and the Planck coincidence reads $\mathcal{D}_3(T_P = 1) = 2 \ln 2$ Planck lengths — the identity in its simplest form.

Extensions to Kerr, Reissner-Nordström, and Kerr-Newman black holes (yielding $R_{\text{total}} = f(\chi)r_s$ and the master formula $h(\chi, q) = 4\sigma/(2(1+\sigma) - q^2)$ [23]), to all spacetime dimensions [25], and to Anti-de Sitter spacetimes with holographic CFT interpretation [26] are presented in companion papers. Together these establish the D3 framework as a unified information-geometry approach to black hole thermodynamics.

ACKNOWLEDGMENTS

The author thanks Dr. Aritra Ghosh (Rochester Institute of Technology) for valuable feedback on the manuscript and for discussions regarding the connection between the D3 framework and area quantisation approaches to black hole thermodynamics, and Prof. Bijan Bagchi for discussions regarding the Kerr extension. This research received no external funding.

Appendix A: Dimensional Analysis

1. SI unit verification

The D3 formula has dimensions:

$$\begin{aligned} [\mathcal{D}_3] &= \frac{[G][k_B][T]}{[c]^4} \\ &= \frac{(\text{m}^3 \text{kg}^{-1} \text{s}^{-2})(\text{kg m}^2 \text{s}^{-2} \text{K}^{-1})(\text{K})}{(\text{m s}^{-1})^4} \\ &= \frac{\text{m}^5 \text{s}^{-4}}{\text{m}^4 \text{s}^{-4}} = \text{m}. \end{aligned}$$

All intermediate units cancel, confirming the formula has dimensions of length.

2. Planck unit form

In Planck units ($G = c = k_B = \hbar = 1$), Eq. (5) reduces to $\mathcal{D}_3 = 2T \ln 2$. At $T = T_P = 1$: $\mathcal{D}_3 = 2 \ln 2$ Planck lengths. The formula has no free parameters in Planck units.

3. Ratio to Schwarzschild radius

At the Hawking temperature $T_H = \hbar c^3 / (8\pi G M k_B)$:

$$\begin{aligned} \frac{\mathcal{D}_3(T_H)}{r_s(M)} &= \frac{2Gk_B T_H \ln 2 / c^4}{2GM/c^2} \\ &= \frac{k_B T_H \ln 2}{Mc^2} = \frac{\hbar c \ln 2}{8\pi G M^2}. \end{aligned} \quad (\text{A1})$$

This equals $(M_{\text{QG}}/M)^2$ [24], where $M_{\text{QG}} = m_P \sqrt{\ln 2 / (8\pi)}$. For $M \gg M_{\text{QG}}$, $\mathcal{D}_3 \ll r_s$: each bit erased shifts the horizon by a negligible fraction of its own size.

Appendix B: Numerical Verification

1. D3 formula at standard temperatures

Using CODATA 2018 constants: $G = 6.67430 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$, $k_B = 1.380649 \times 10^{-23} \text{ J K}^{-1}$, $c = 2.99792458 \times 10^8 \text{ m s}^{-1}$:

$$\mathcal{D}_3(300 \text{ K}) = 4.744 \times 10^{-65} \text{ m/bit},$$

$$\mathcal{D}_3(310 \text{ K}) = 4.903 \times 10^{-65} \text{ m/bit}.$$

2. Planck coincidence verification

Using also $\hbar = 1.054571817 \times 10^{-34} \text{ J s}$:

$$T_P = \sqrt{\hbar c^5 / (G k_B^2)} = 1.416784 \times 10^{32} \text{ K},$$

$$\ell_P = \sqrt{\hbar G / c^3} = 1.616255 \times 10^{-35} \text{ m},$$

$$\mathcal{D}_3(T_P) / \ell_P = 1.386294 \dots = 2 \ln 2.$$

Agreement to six decimal places. The algebraic proof (Sec. III) guarantees this to arbitrary precision.

All verification code is available at <https://github.com/bsharma173860-oss/d3-research>.

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